zp12 *sampling drums*

The keys



White 1–8	pads of bank **A** (or **C**)
White 9–16	pads of bank **B** (or **D**)
OCT+ **OCT−**	banks **C + D** / back to **A + B** (the LEDs A–D show which)
black keys	**loops 1–11** (12–16: SEQ → LOOP): stopped at once; playing: from the end of the loop (its key blinks, the LCD says **LOOP 3 NEXT**). The loop playing is lit.
ARP	**MULTI PITCH**: the last sound on all 27 keys, F4 (white 8) as tuned. ARP again: pads.
USB MIDI	notes 36–67 play pads A1–D8; Program Change 0–15 picks loop 1–16 (at the loop's end)

The factory sounds

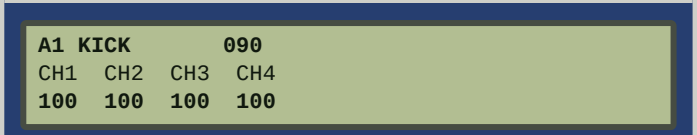
A1 KICK	**A2** SNARE	**A3** HAT	**A4** CLAP	**A5** TOM H	**A6** CRASH	**A7** BASS	**A8** PNO Cm9
B1 KICK 33	**B2** SNARE 33	**B3** OHAT	**B4** CLAP 33	**B5** CONGA	**B6** RIDE	**B7** SCRCH	**B8** PNO F13
C1 TOM L	**C2** RIM 33	**C3** HAT 33	**C4** SNAP	**C5** CLAVE	**C6** COWBL	**C7** WOOD	**C8** EP Dm9
D1 SPIN	**D2** RIM	**D3** SHAKR	**D4** PIANO	**D5** VIBES	**D6** HORNS	**D7** TAMB	**D8** EP Gm9

33: TUNE −7 + SPEED 33, about an octave lower. Any pad, any of these 27 samples or your own (the sample editor): hit it, turn **PRESETS**. 12 bit at 26.04 kHz, pitched without interpolation.

Sound

Channels	a pad plays on the channel of its **position**: pad 3 of every bank on channel 3 (fader 3, mute 3, solo 3). One sound at a time each: a new hit cuts the last (the hats share 3); a CHOP group (EDIT → OUT) cuts across channels. Pads 1–2 have the filter
1–2	a 4-pole resonant low-pass: open as CUT at the hit, closing two octaves as the sound decays (RESO)
3–6 · 7–8	a fixed low-pass · no filter (OUT: CUT / RESO --)
45 → 33	TRUNC, SPEED: played at 33/45 of its rate: lower, grainier. Own samples also at ×2 (the editor): an octave down on the pad

The knobs



KNOB 1–4	the **faders** of channels 1–4 (100 as set, 50 −12 dB); **SEL**: **5–8** (lit) / 1–4, also from a page. The LCD's four columns always say what the knobs do.
A page	EDIT, FX, SEQ, GLO open one (tap again: the next); untouched **12 s** (SETUP: BACK), or **HOME**: the faders again
SELECT	tempo
ALGORITHM	the sound to edit (A1 … D8), without playing it
PRESETS	**the sample** of the selected pad (the last hit, or ALGORITHM): one sample on many pads, each with its own TUNE, DECAY …
MASTER	the volume

Pages · KNOB 1 · 2 · 3 · 4

EDIT WAVE	the pad's sample · COPY> pad · COPY (turn twice) · DELETE its own sample (turn twice)
SOUND	TUNE (semitones) · FINE · DECAY · LEVEL
TRUNC	START · END · DIR (fwd / rev) · SPEED (45 / 33)
OUT	CHOP group (--, 1–8: its pads cut each other, wherever they are; hold pads while turning: they all join; the label shows the channel = the pad's position) · PAN · CUT · RESO (only pads 1–2 filter: on 3–8 they show --)
SENDS	DRIVE · CHO · DLY · REV: **the effects of this pad**
FX FILTER	**DJ filter** on the whole mix: left a low-pass, right a high-pass, centre OFF · RESO
CHORUS	RATE · DEPTH · MIX
DELAY	TIME (1/4 … 16T, in tempo) · FDBK · COLOR · MIX
REVERB	SIZE · DAMP · PRE (ms)
SEQ LOOP	LOOP · BARS (AUTO, 1–32) · QUANT (AUTO CORRECT) · SWING 50–71 %
LOOP TOOLS	CLEAR (turn twice) · COPY> loop · COPY (turn twice)
SONG	STEP · SEG (its loop) · REPEAT (0: the song ends here) · SONG OFF / 1–4 (stopped)
GLO SETUP	TEMPO · BACK · – · RESET (**factory: all loops gone**, turn twice); tap: **CLICK**: CLICK · **COUNT** · **DUB** (NOW / BAR); tap: OUTPUT: SMOOTH; tap: **SAMPLE**: RATE · TIME · THRESH · MON

Make a beat

1. **REC** (**ARMED**), **PLAY**: **COUNT 4 3 2 1**, then it records.
2. Play kick and snare. **AUTO CORRECT** puts them on the grid.
3. **REC** again at the end: an empty loop (BARS **AUTO**) is now as long as your take.
4. Hats step by step: hit the hat, **hold** **SEQ**: the white keys are its 16 steps (lit = a hit), tap to set / clear; **OCT+** / **OCT−** the next / last bar.
5. **REC** while it plays: overdub on top (DUB BAR: from the next 1); **REC** again: off.
6. Turn TUNE, FINE, DECAY (EDIT → SOUND) or CUT while recording: **the pad's hits keep what you turned** (a red dot in the step grid).
7. A chord: white 8 (A8), or **OCT+**, white 8 (C8). Melodic: **ARP** (MULTI PITCH).
8. Another loop: a black key, record into it. Songs: SEQ → SONG.

Buttons

PLAY	run / stop (lit while it runs)
REC	stopped: ARMED; playing: overdub on / off; **hold 2 s**: clear the loop
GLO held + white	1–8 **mute**, 9–16 **solo** ch 1–8 (lit while heard; M / S under the faders)
EDIT held + OCT−	**undo** the last clear / erase / copy (again: redo)
FX held + white 1–4	**punch-in** while held: 1 roll 1/8 · 2 roll 1/16 · 3 reverse · 4 tape stop; **OCT−** / **OCT+**: **reverb / echo throw** (FX tapped: the FX pages)
LFO + pad	**erase** its hits as the playhead passes (stopped: all); **LFO** tapped: **SAMPLE**
ENV	tap tempo (it blinks on the beat)
SAVE	save now (playing: at the stop; also by itself); **held + a black key**: the loop into that loop
HOME	the faders

Effects in 3 steps

1. Hit the pad (the snare).
2. **EDIT** until the LCD says **SENDS**.
3. KNOB 4 = REV, KNOB 3 = DLY. Shape them: **FX**.

Your start: it plays A5, C1, A1 + A6: put your own sounds there. **In the browser** (Chrome / Edge, USB): **zp12.designburgapps.com/editor** samples (mic, interface, line-in), drops your own sounds onto the pads or **chops a loop to pads** (one chop a pad), **/backup** saves and restores everything. No answer? Hold OCT− while switching on (USB rescue).

Based on SLOOP (isod89) and Felucca (Leo Kuroshita, Hügelton Instruments), GPL-3.0 · sounds CC0 (VSCO-2 CE, VCSL, Sonic Pi), E-piano Greg Sullivan (CC BY 3.0)

Inspired by the 12-bit samplers of the 80s, names their owners' · not affiliated with M-VAVE

What the lights say

PLAY	lit while it runs
REC	lit: recording · blinking: ARMED, or an overdub waiting for the next 1
ENV	blinks on the beat (tap along)
ARP · SEL	MULTI PITCH on · the knobs are faders 5–8
LFO	erasing (held + a pad)
SAVE	lit in song mode
OCT− OCT+	banks A + B / C + D on the keys
HOME …	the button of the open page (HOME, EDIT, FX, SEQ, GLO)
White keys	flash when their pad sounds (played or from the loop); **SEQ held**: the steps with a hit, the playhead glowing; **GLO held**: 1–8 the channels you hear, 9–16 the soloed
Black keys	the loop playing lit, the next one blinking, the others glowing

The screen

Big line	the selected pad and its sound (A1 KICK), or what just changed
Top right	the tempo, with **ARMED**, **COUNT** or **REC** before it; below it the page
Four columns	what KNOB 1–4 do now, and their values
A B C D	the banks on the keys
M · S	MULTI PITCH · song mode
RUN · REC	running (blinks on the beat) · recording (blinks: ARMED)
M / S	under a fader: that channel muted / soloed
Red dot	in the step grid: a hit with its own TUNE / FINE / DECAY / CUT
Version	on the GLO pages and at the start (any key skips the start)

Sampling over USB (2.5)

1. On the computer (or a phone with USB-C) choose **zp12 FM-1** as the sound output (Linux may call it "pid.codes Test PID"), volume up, play the sound.
2. Hit the pad it should go on. LFO tapped: **SAMPLE**, the meter moves.
3. RATE · TIME · THRESH (OFF: at once) · MON (hear it). REC: ERASE %, then REC; REC again or TIME ends it.

On the pad as SMP01–24; again on that pad: replaced. EDIT → WAVE → DELETE frees it. Room ~1.8 s at 26 kHz (more with the bank room, the editor allows it).

Recording

Count-in	GLO → CLICK → COUNT: OFF (it records at once), 1 bar (4 3 2 1), 2 bars (8 … 1)
AUTO	a loop with BARS AUTO takes the length of its first take, in whole bars (REC up to a quarter past a bar line: that bar)
QUANT	AUTO CORRECT to OFF, 1/8, 1/8T, **1/16**, 1/16T, 1/32, 1/32T
SWING	50, 54, 58, 63, 67, 71 %: the off 1/16s (with QUANT 1/8: the off 1/8s) come later
Overdub	a hit where one is: the louder stays. Up to 512 hits a loop
Velocity	the keys play at one level; USB MIDI velocity is played and recorded (8 steps)
Per hit	TUNE, FINE, DECAY, CUT turned while recording (within 0.6 s of the turn) stay with the pad's hits
Step grid	16 steps of 1/16 a bar; a step set by hand plays at a fixed level. A hit off the 1/16 grid shows, but a tap adds one beside it. In an AUTO loop the first step makes it 1 bar
Erase	LFO + a pad: playing, its hits go as the playhead passes; stopped, all at once

Loops and songs

16 loops	1–11 on the black keys, 12–16 with SEQ → LOOP (stopped); each 1–32 bars or AUTO
Copy	SAVE held + a black key, or SEQ → TOOLS: COPY> the loop, COPY (turn twice)
Songs	4 songs of up to 32 steps, each step a loop (SEG) played REPEAT times
Make one	stopped, SEQ → SONG: SONG 1–4, then STEP and SEG; turn SEG on the step after the last to add one; REPEAT to 0 to end it there
Play	PLAY plays the song from step 1 and stops at its end. A black key or SONG OFF: back to the loops

Ranges

TEMPO	40–240 BPM; SELECT 0.5 a click, ENV tapped
Faders	0–127: 100 as set, 50 −12 dB, 127 about +4 dB
TUNE · FINE	−24 … +12 semitones · ±50 cents
DELAY	TIME 1/4, 1/8, 1/16, 1/32, 8T, 16T; FDBK 0–120
REVERB PRE	0–90 ms
Filters	pads 1–2: CUT, RESO (4-pole low-pass) · 3–4: 9 kHz · 5–6: 12 kHz · 7–8: none
OUT CH1–8	the label shows the channel: it is the pad's position and does not change. Chops that should cut each other: a CHOP group

Saving and safety

Saved	by itself when stopped and 3 s after a change; SAVE now (playing: SAVES AT STOP)
Not saved	mute / solo and the DJ filter (FX → FILTER): gone at power-off
Undo	EDIT held + OCT−: the last clear, copy or stopped erase of a loop, one step. Not an erase while playing, not WAVE → COPY
RESET	GLO → SETUP: every pad, mix, effect back to the factory, **all loops and songs gone**. Back up first
Backup	**zp12.designburgapps.com/backup**: everything to a file and back (Chrome / Edge, USB)
Rescue	no answer? Hold OCT− while switching on: the USB rescue, then install again
DJ filter	sounds dull or thin? FX → FILTER may be on (centre = OFF)

GPL-3.0 · github.com/zvenson/zp12